

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Course: B.Tech Branch : Artificial Intelligence & Data Science / Artificial

Intelligence(AI) & Data Science

Semester :IV

Subject Code & Name: BTBS404 Probability Theory and Random Process

Max Marks: 60

Date: 27/12/2024

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/C O)	Mar ks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Events are said to be _____ if the happening of any one of them precludes the happening of all others. a. Independent Event b. Dependent Event c. Mutually Exclusive Event d. Multiplicative event	CO1	1
2	What is the primary objective of estimation in statistics? a. To describe data b. To make predictions about future data c. To determine the value of a population parameter based on sample data d. To test hypotheses	CO5	1
3	_____ provide a unique approach through which outcome of an experiment can be expressed in terms of numbers. a. Constant b. Numerical value c. Independent Variable d. Random Variable	CO5	1

4	For expectation, if X and Y are independent variables then the value of $E(XY)$ is				CO2	1				
	a. $E(X).E(Y)$	b. $E(X)-E(X)$	c. $E(X+Y)$	d. $E(X)-E(Y)$						
5	If the probability density function of a variable X is:					CO2	1			
	X	0	1	2	3			4	5	6
	P(X)	K	3k	5k	7k			9k	11k	13k
	With $k=1/49$ then $P(X \leq 4)$ is									
a. 1		b. $4/49$		c. $25/49$		d. -1				
6	Assuming that a box contains in all ten balls of which six are yellow and four are blue, what is probability of drawing a blue ball?					CO1	1			
	a. 0.1		b. 0.4		c. 0.5			d. 1		
7	Find the mean and variance of the function X					CO2				
	X	1	3	4	5					
	$f_x(X=x)$	0.4	0.1	0.2	0.3					
	a. Mean=3, variance=3							b. Mean= 3, variance=4		c. Mean=4, variance=4
8	Regression coefficients are affected by:					CO4	1			
	a. Change in origin and scale		b. Change in origin only		c. Change in scale only			d. Neither origin nor scale		
9	What does correlation measure					CO3	1			
	a. Cause and effect relationship		b. Degree of association between two variables		c. Statistical independence			d. None of the above		
10	When both variables move in the same direction, the correlation is called					CO3	1			
	a. Zero correlation		b. Negative correlation		c. Positive correlation			d. No correlation		
11	The sample size required for a study is inversely proportional to:					CO3	1			

	a. The level of confidence	b. The margin of error	c. The variability in the population	d. The standard deviation		
12	The sign of regression coefficients and the correlation coefficient is				CO4	1
	a. Always the same	b. Always different	c. Independent of each other	d. Opposite in sign		
Q.2	Solve the following.					12
A)	An insurance company insured 2000 scooter drivers, 4000 car drivers and 6000 truck drivers. The probability of accident are 0.01,0.03 and 0.15 for scooter driver, car driver and truck driver respectively. One of the insured person meets with an accident. What is the Probability that he is a scooter driver ?				CO1	6
B)	There are 500 students in a certain school. 150 students are enrolled in an Algebra course and 80 students are enrolled in a Chemistry course. There are 30 students who are taking both Algebra and Chemistry. If a student is chosen at random a) What is the probability that student is taking Algebra? b) What is the probability that student is taking Chemistry, given that the student is also taking Algebra? c) What is the probability that student is taking Algebra , given that the student is also taking Chemistry?				CO1	6
Q.3	Solve the following.					12
A)	The average number of acres burned by forest and range fires in a large New Mexico county is 4,300 acres per year, with a standard deviation of 750 acres. The distribution of the number of acres burned is normal. What is the probability that between 2,500 and 4,200 acres will be burned in any given year? Given $Z(0.13) = 0.0082$ $Z(2.40) = 0.4483$				CO2	6
B)	In a factory of blades, there is a chance of 1/500 for any blade to be defective. Blades are supplied in a packet of 10. By using Poisson distribution calculate approximate number of packets containing no defective and two defective blades in 10000 packets.				CO2	6
Q.4	Solve Any Two of the following.					12
A)	$\sum xy=70$, $r=0.9$, $\sigma_y=3.5$ $\sum x^2=100$ Find number of items if x and y are derivations of arithmetic mean				CO3	6

B)	Calculate the Spearman's rank correlation coefficient for the following data.							CO3	6																		
	<table><tr><td>Candidates</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>Marks in Tamil</td><td>75</td><td>40</td><td>52</td><td>65</td><td>60</td></tr><tr><td>Marks in English</td><td>25</td><td>42</td><td>35</td><td>29</td><td>33</td></tr></table>							Candidates	1	2	3	4	5	Marks in Tamil	75	40	52	65	60	Marks in English	25	42	35	29	33		
Candidates	1	2	3	4	5																						
Marks in Tamil	75	40	52	65	60																						
Marks in English	25	42	35	29	33																						
C)	Compute Karl Pearson's coefficient of correlation from the following data:							CO3	6																		
	<table><tr><td>Density (per sq. km)</td><td>2000</td><td>5000</td><td>4000</td><td>7000</td><td>6000</td><td>3000</td></tr><tr><td>Patients of dengue fever</td><td>100</td><td>160</td><td>140</td><td>200</td><td>170</td><td>130</td></tr></table>							Density (per sq. km)	2000	5000	4000	7000	6000	3000	Patients of dengue fever	100	160	140	200	170	130						
Density (per sq. km)	2000	5000	4000	7000	6000	3000																					
Patients of dengue fever	100	160	140	200	170	130																					
Q.5	Solve Any Two of the following.								12																		
A)	If two lines of regression are $9x+y-\lambda=0$ and $4x+y=\mu$ and means of x and y are 2 and -3 respectively. Find the values of μ and λ and coefficient of correlation between x and y. Also find b_{yx} and b_{xy}							CO4	6																		
B)	Calculate regression equation of x on y and y on x from the following data and estimate x when $y=26$ and estimate y when $x=21$							CO4	6																		
	10	12	13	17	18	20	24	30																			
	5	6	7	9	13	15	20	21																			
C)	Explain Theorem on regression coefficient and give properties of regression coefficient							CO4	6																		
Q. 6	Solve Any Two of the following.								12																		
A)	Explain the concept of hypothesis testing in statistics. Discuss the steps involved in conducting a hypothesis test and the importance of setting a significance level.							CO5	6																		
B)	What are the assumptions required for small sample estimation of a population mean? Discuss how violating these assumptions affects the results.							CO5	6																		
C)	Explain the process of estimating a population mean using a large sample							CO5	6																		
*** End ***																											